

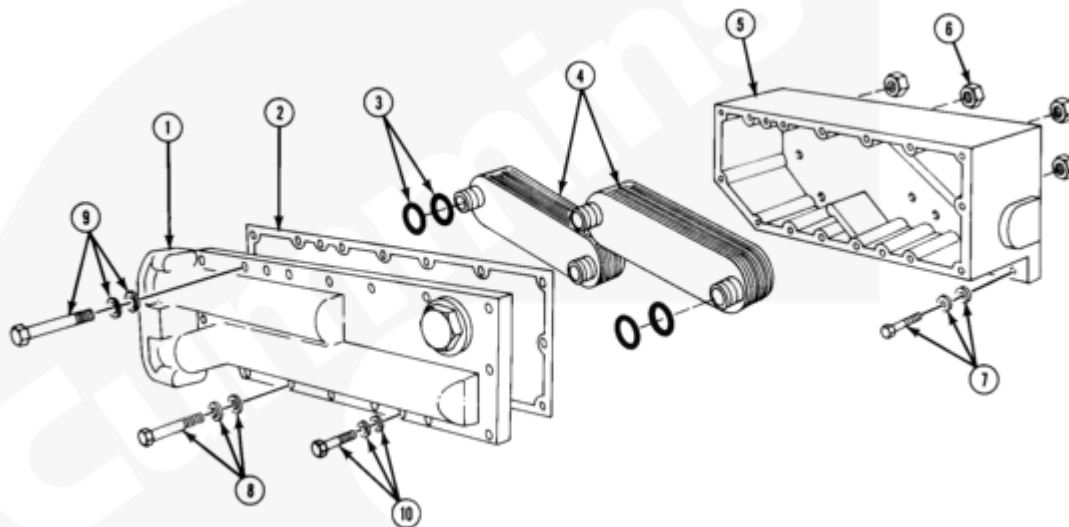
Trainings ▾

General Information

The engine mounted torque converter cooler is available as an **option**. This **option** can be added to a **standard** engine by purchasing the torque converter cooler housing, the cooler cover, and the elements.

The location of the turbocharger oil drain line, turbocharger water supply line, and the other components will have to be changed on some engines. The torque converter cooler is mounted and **replaces** the lubricating oil cooler cover. When the torque converter cooler **option** is selected, longer capscrews **must** be used. With the torque converter cooler option, the turbocharger water supply line **must** also be moved from the oil cooler cover to a fitting in the top of the torque converter cooler housing.

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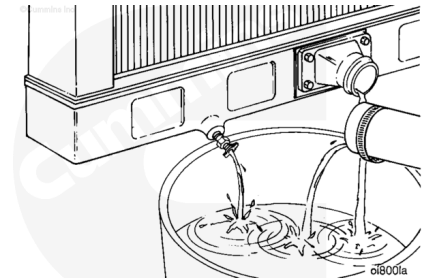
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1. Torque converter oil cooler cover
2. Torque converter oil cooler cover gasket
3. O-ring
4. Torque converter oil cooler element
5. Torque converter oil cooler housing
6. Self-locking nut
7. Capscrew, lock washer, and plain washer
8. Capscrew and lock washer
9. Capscrew, lock washer, and plain washer
10. Capscrew, lock washer, and plain washer.

Preparatory Steps

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.



⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

⚠ CAUTION ⚠

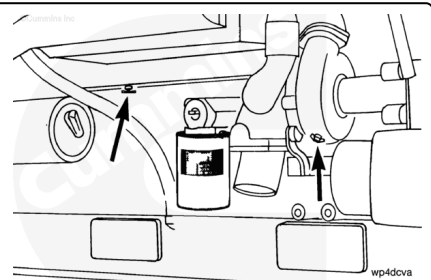
Use caution when draining coolant that the coolant is not spilled or drained into the bilge area. Do not pump coolant overboard. If the coolant is not reused, it must be disposed of in accordance with local environmental regulations.

Note : Shown in the following steps is the QSK19 engine mechanically actuated injectors. Although there is some difference, the basic procedure remains the same.

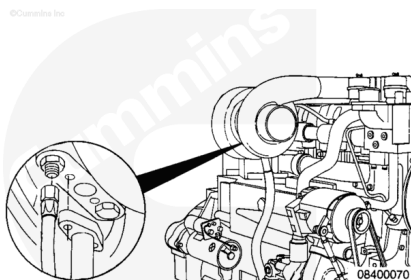
Drain the cooling system. Refer to Procedure 008-018 (</qs3/pubsys2/xml/en/procedures/20/20-008-018-tr.html>).

On marine applications, **open** the draincocks on the water pump and oil cooler housing.

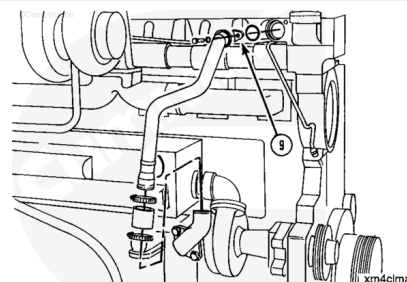
Close the draincock(s) after all the coolant is removed.



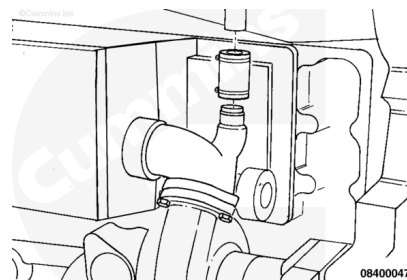
Remove the turbocharger oil drain hose. Remove the turbocharger coolant supply hose.



Remove the water bypass tube clamp (9). Loosen both hose clamps. Remove the bypass tube and discard the o-ring.

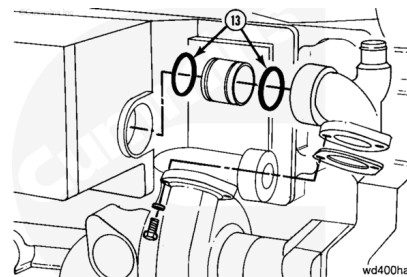


Disconnect the aftercooler supply hose from the water pump outlet connection.

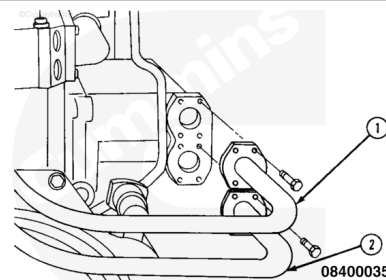


Remove the water pump outlet connection assembly.

Remove the water transfer tube from the water outlet connection. Remove and discard the two o-rings (13) and the gasket.



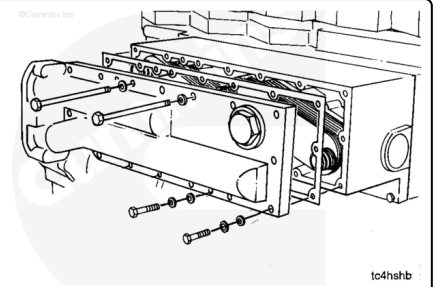
Remove the oil supply (1) and oil return (2) hoses from the torque converter cooler.



Remove

Remove the torque converter cooler cover mounting capscrews.

Note : The cover must be pried from the housing because of the tight fit between the cover and the o-rings on the elements.



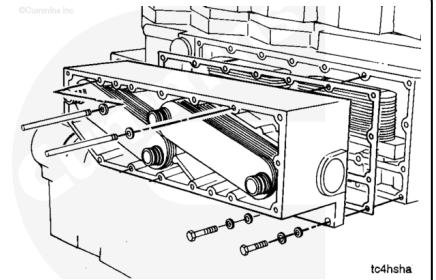
⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

Use two [3/8-16 x 12 in] guide studs to support the housing while the capscrews are being removed.

Remove the remaining capscrews, the torque converter cooler housing, and the gasket.

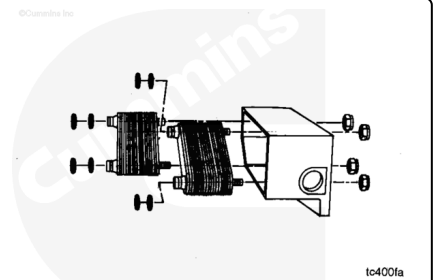
Discard the gasket.



Remove the torque converter cooler element mounting nuts.

Remove the elements.

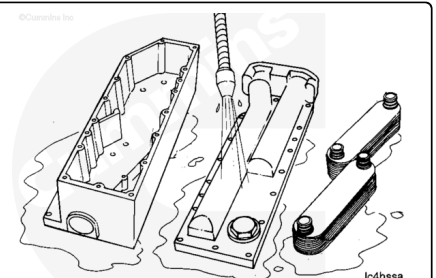
Remove and discard the o-rings.



Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



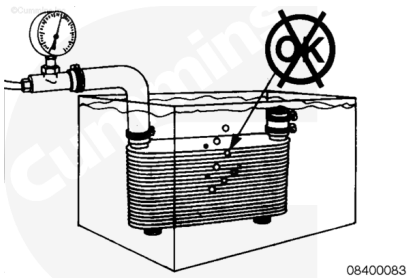
Use a solvent that will **not** harm aluminum. Clean the parts.

Inspect the parts for cracks or other damage.

Pressure Test

Note : Heating the water in the tank to 50°C [120°F] will improve the test results.

Pressure test the elements to check for leaks.



Measurements

	kpa	psi
Air Pressure	415	60

Note : Discard the elements if any debris is found, the engine has had a debris causing failure, or the torque converter has failed.

Discard the elements if any leaks are present.

Install

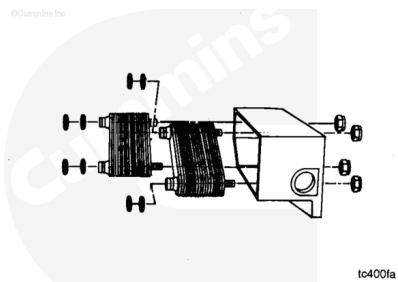
Install the elements.

Tighten the self-locking nuts.

Torque Value: 130 n·m [95 ft-lb]

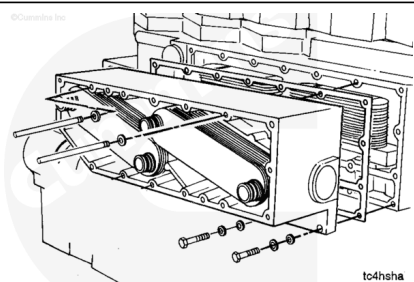
Use new o-rings. Lubricate the o-rings with vegetable oil.

Install the o-rings on the cooler elements.



⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



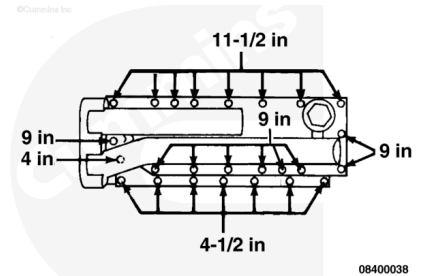
Use two [3/8-16 x 12 inch] guide studs to support the housing during installation.

Install the gasket and housing.

Install the capscrews in the bottom row of holes on the housing. These capscrews are all 101.6 mm [4 in] long.

Note : The length and location of all the capscrews on the torque converter cooler and cover are shown. Be sure the capscrews are the correct length.

Do **not** tighten any of the capscrews until all of them have been installed.

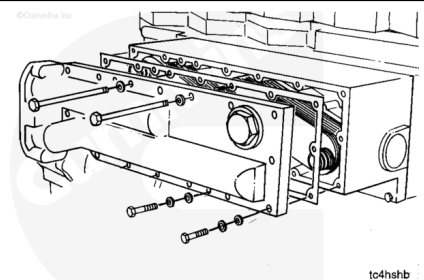


Note : Be sure the capscrews are the correct length.

Use vegetable oil to lubricate the o-rings on the cooler element and the bores in the cover.

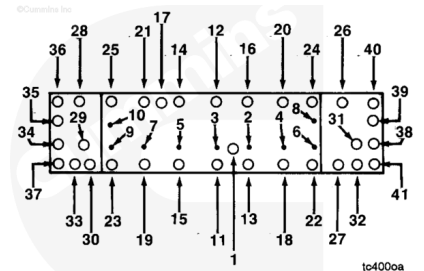
Install the gasket and cover. Push the cover over the o-rings until the cover is against the housing.

Install the remaining capscrews.



Tighten the capscrews in the sequence shown.

Torque Value: 45 n•m [35 ft-lb]



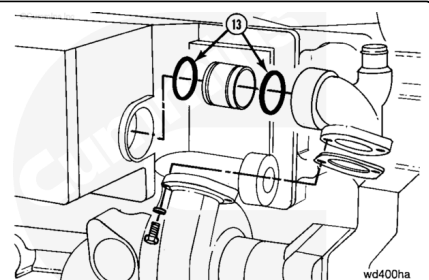
Finishing Steps

Use vegetable oil. Lubricate the o-rings (13).

Install the transfer tube in the water pump outlet connection.

Install the water pump outlet connection, gasket, and capscrews.

Tighten the capscrews.

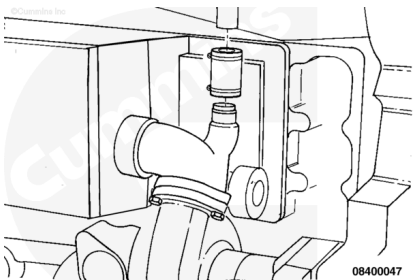


Torque Value: 45 n·m [35 ft-lb]

Connect the aftercooler supply hose to the water pump outlet connection.

Tighten the clamps.

Torque Value: 5.6 n·m [50 in-lb]

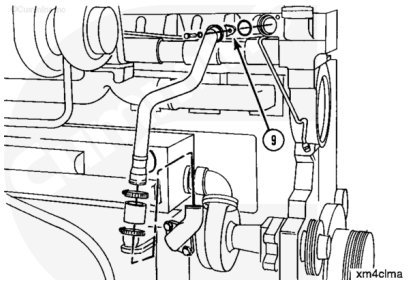


Use vegetable oil. Lubricate the o-ring on the water bypass tube.

Install the bypass tube.

Install the retainer (9) and capscrew.

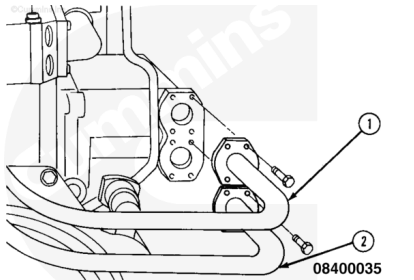
Tighten the capscrew and hose clamp.



Capscrew	45 n.m	[35 ft-lb]
Clamp	5.6 n.m	[50 in-lb]

Connect the torque converter oil hoses to the torque converter cooler.

Capscrew	45 n.m	[35 ft-lb]
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⚠ CAUTION ⚠

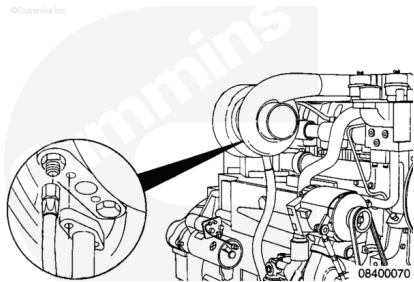
These parts are safety-related items. Be sure the parts are clamped correctly to prevent fretting.

Install the turbocharger oil drain tube.

Tighten the capscrews.

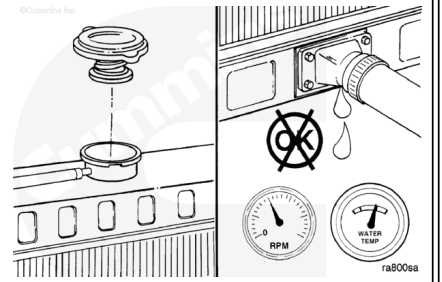
Torque Value: 45 n·m [33 ft-lb]

Connect the coolant supply hose to the turbocharger.



⚠ CAUTION ⚠

Use caution when filling coolant that the coolant is not spilled into the bilge area. Do not pump coolant overboard. If the coolant is not reused, it must be disposed of in accordance with local environmental regulations.



Fill the cooling system. Refer to Procedure 008-018 (</qs3/pubsys2/xml/en/procedures/20/20-008-018-tr.html>).

Operate the engine at 70°C [160°F] coolant temperature.

Check for leaks.